Sequences

- A list of numbers
  ↳ $a_0, a_1, a_2, a_3, \dots$

Example:

$0, 1, 2, 3, \dots \Longrightarrow n$, where $n \in \mathbb{Z}$

$1, 1/2, 1/4, 1/16, \dots \Longrightarrow \frac{1}{2^n}$, where $n \in \mathbb{Z}$

$1, -1, 1, -1, \dots \Longrightarrow (-1)^n$, where $n \in \mathbb{Z}$

A sequence converges if the elements approach a finite value as n runs from 0 to ∞.

A finite value

$$\boxed{\lim_{n \to \infty} a_n = L}$$

If this is not sastisfied, the sequence diverges

Important Limit

$$\lim_{n \to \infty} \left(1 + \frac{a}{n}\right)^n = e^a$$

Series

- Sum of a sequence
  ↳ $\sum_{n=0}^{\infty} a_n = a_0 + a_1 + a_2 + \dots$

Example:

$1 + 1/2 + 1/3 + 1/4 + \dots + 1/n \Longrightarrow \sum_{n=0}^{\infty} \frac{1}{n}$ [Harmonic Series]

$1 + 1/2 + 1/4 + 1/8 + \dots + \frac{1}{2^n} \Longrightarrow \sum_{n=0}^{\infty} \frac{1}{2^n}$

$1 + (-1) + 1 + (-1) + \dots + (-1)^n \Longrightarrow \sum_{n=0}^{\infty} (-1)^n$

Geometric Series $\sum_{n=0}^{\infty} a \cdot r^n$

A series converges if the sum gets closer to a finite value

↳ That means the series must add smaller & smaller numbers [Sequence must converge to 0]

$$\boxed{\lim_{n \to \infty} a_n = 0}$$

If this is not sastistied, the series diverges

Take Note:

For a series to converge, its sequence must converge to 0

However, a sequence converging to 0 does not guarantee the series converges [Eg. Harmonic Series]

Ratio Test

Since $\lim_{n\to\infty} a_n = 0$ does not guarantee that the series converges, we use the ratio test.

$$\boxed{\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L}$$

① $L < 1$, series converges

② $L > 1$, series diverges

③ $L = 1$, inconclusive

Recall Geometric Series :

$$\sum_{n=1}^{\infty} \underbrace{ar^{n-1}}_{a_n} = \begin{cases} \dfrac{a}{1-r} & |r| < 1 \\ diverges & |r| \geq 1 \end{cases}$$

$$\frac{a_{n+1}}{a_n} = \frac{ar^n}{ar^{n-1}} = r$$

constant

Power Series

$$\boxed{\sum_{n=0}^{\infty} c_n (x-a)^n}$$

a sequence

Geometric Series:

$$\sum_{n=0}^{\infty} x^n = \begin{cases} \dfrac{1}{1-x} & |x| < 1 \\ diverges & |x| \geq 1 \end{cases}$$

can take any values btwn -1 and 1

Example 1 :

$$\sum_{n=0}^{\infty} n!\, x^n$$

Using ratio test,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)!\, x^{n+1}}{n!\, x^n}\right|$$

$$= \lim_{n\to\infty} \left|(n+1)x\right| = \begin{cases} 0, & x = 0 \\ \infty, & x \neq 0 \end{cases}$$

∴ For series to converge,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$$

$$\lim_{n\to\infty} \left|(n+1)x\right| < 1$$

Since as $n\to\infty$, $(n+1)\to\infty$. The only way for $\lim_{n\to\infty} \left|(n+1)x\right| < 1$ is when $x = 0$ #

∴ Series converges when $x = 0$

∴ For series to diverge,

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$$

$$\lim_{n\to\infty} \left|(n+1)x\right| > 1$$

Since as $n\to\infty$, $(n+1)\to\infty$.

$\lim_{n\to\infty} \left|(n+1)x\right| > 1$ when $x \neq 0$

∴ Series diverges when $x \neq 0$

Example 2:

$$\sum_{n=0}^{\infty} \frac{(3x)^n}{n^8}$$

Using ratio test,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(3x)^{n+1}}{(n+1)^8}\cdot\frac{n^8}{(3x)^n}\right|$$

$$= |3x| \lim_{n\to\infty}\left|\frac{1}{(1+\frac{1}{n})^8}\right|$$

$$= |3x|$$

$\therefore$ For series to converge,

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$$

$$|3x| < 1$$

$$|x - \underbrace{0}_{\text{centre}}| < \underbrace{\frac{1}{3}}_{\text{radius}}$$

$$\Longrightarrow -\frac{1}{3} < x < \frac{1}{3}$$

$\therefore$ Series converges at $-\frac{1}{3} < x < \frac{1}{3}$.

[Radius of convergence is $\frac{1}{3}$]

Power Series of solution of ODEs

Example 1:

$$y' = 1 - 3t + y + t^2 + ty \quad , \text{ with IC } y(0) = 0$$

Let $y = \sum_{n=0}^{\infty} c_n t^n = c_0 + c_1 t + c_2 t^2 + \ldots$ —— ①

$y' = c_1 + 2c_2 t + 3c_3 t^2 + \ldots + n c_n t^{n-1}$ —— ②

Using IC, $y(0) = c_0 = 0$ —— ③

Subsituiting ① & ② into ODE,

$$c_1 + 2c_2 t + 3c_3 t^2 + 4c_4 t^3 + \ldots = 1 - 3t + [c_0 + c_1 t + c_2 t^2 + c_3 t^3 + \ldots]$$
$$+ t^2 + [c_0 + c_1 t + c_2 t^2 + c_3 t^3 + \ldots]t$$

$\vdots$

$c_1 = 1 + c_0 = 1$ * [using ③]

$2c_2 = -3 + c_1 + c_0$

$c_2 = -1$ * [using ③]

$3c_3 = c_2 + 1 + c_1$

$c_3 = 1/3$ * [using ③]

$\therefore \; y = t - t^2 + \frac{1}{3}t^3 + \ldots$

[This is a very good approx. for small t.]

so that not included terms don't affect y as much.

<u>3.3: MACLAURIN'S SERIES</u>

<u>Finding the Maclaurin's Series from First Principles</u>

1. The Maclaurin's series for an infinitely differentiable function $f(x)$ about the point $x = 0$, is given by

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

In summation form this is written as

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

where $f^{(n)}(0)$ is the nth derivative evaluated at the point $x = 0$.

2. ***Activity***:
Find the Maclaurin's series for the following functions, giving the first four non-zero terms.
(a) $f(x) = e^x$
(b) $f(x) = \ln(1 + x)$
(c) $f(x) = \sin x$
(d) $f(x) = \cos x$
(e) $f(x) = \tan^{-1} x$
(f) $f(x) = \sinh x$
(g) $f(x) = \cosh x$
(h) $f(x) = \tanh^{-1} x$
Check your answers by referring to the standard results listed on Page 5 of the Formula Booklet (MF19).

Example 8

If $f(x) = \sin^{-1} x$, find the first three non-zero terms of the Maclaurin's series of this function.

$\left(Ans: \sin^{-1} x \approx x + \frac{1}{6}x^3 + \frac{3}{40}x^5\right)$

Find also an estimate, to 9 decimal places, for $\sin^{-1} 0.1$. $(Ans: 0.100167417)$

Solution

$f(x) = \sin^{-1} x$ $\quad f(0) = 0$

$f'(x) = \dfrac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}$ $\quad f'(0) = 1$

$f''(x) = -\frac{1}{2}(1-x^2)^{-3/2}(-2x)$ $\quad f''(0) = 0$

$= x(1-x^2)^{-3/2}$

$f'''(x) = x\left(-\frac{3}{2}\right)(1-x^2)^{-5/2}(-2x) + (1-x^2)^{-3/2}$ $\quad f'''(0) = 1$

$= 3x^2(1-x^2)^{-5/2} + (1-x^2)^{-3/2}$

$f^{(4)}(x) = 3x^2\left(-\frac{5}{2}\right)(1-x^2)^{-7/2}(-2x) + 6x(1-x^2)^{-5/2} + \left(-\frac{3}{2}\right)(1-x^2)^{-5/2}(-2x)$

$= 15x^3(1-x^2)^{-7/2} + 9x(1-x^2)^{-5/2}$ $\quad f^{(4)}(0) = 0$

$f^{(5)}(x) = 15x^3\left(-\frac{7}{2}\right)(1-x^2)^{-9/2}(-2x) + 45x^2(1-x^2)^{-7/2} + 9x\left(-\frac{5}{2}\right)(1-x^2)^{-7/2}(-2x) + 9(1-x^2)^{-5/2}$

$= 105x^4(1-x^2)^{-9/2} + 45x^2(1-x^2)^{-7/2} + 45x^2(1-x^2)^{-7/2} + 9(1-x^2)^{-5/2}$

$= 105x^4(1-x^2)^{-9/2} + 90x^2(1-x^2)^{-7/2} + 9(1-x^2)^{-5/2}$ $\quad f^{(5)}(0) = 9$

$f(x) = 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 + \frac{1}{3!}x^3 + \frac{0}{4!}x^4 + \frac{9}{5!}x^5 + \ldots$

$= x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \ldots$

$f(x) = \sin^{-1} x \approx x + \frac{1}{6}(0.1)^3 + \frac{3}{40}(0.1)^5$

$= 0.100167417$

2. ***Activity***:

Find the Maclaurin's series for the following functions, giving the first four non-zero terms.

(a) $f(x) = e^x$

(b) $f(x) = \ln(1 + x)$

(c) $f(x) = \sin x$

(d) $f(x) = \cos x$

(e) $f(x) = \tan^{-1} x$

(f) $f(x) = \sinh x$

(g) $f(x) = \cosh x$

(h) $f(x) = \tanh^{-1} x$

Check your answers by referring to the standard results listed on Page 5 of the Formula Booklet (MF19).

b) $f'(x) = \frac{1}{1+x} = (1+x)^{-1} \Rightarrow f'(0) = 1$

$f''(x) = -1(1+x)^{-2} \Rightarrow f''(0) = -1$

$f'''(x) = 2(1+x)^{-3} \Rightarrow f'''(0) = 2$

$f^{(4)}(x) = -6(1+x)^{-4} \Rightarrow f^{(4)}(0) = -6$

∴ Maclauren series:

$$f(x) = 0 + \frac{1}{1!}x + \frac{(-1)}{2!}x^2 + \frac{2}{3!}x^3 + \frac{(-6)}{4!}x^4 + \dots$$

$$= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

c) $f(x) = \sin x \qquad f(0) = 0$

$f'(x) = \cos x \qquad f'(0) = 1$

$f''(x) = -\sin x \qquad f''(0) = 0$

$f'''(x) = -\cos x \qquad f'''(0) = -1$

$f^{(4)}(x) = \sin x \qquad f^{(4)}(0) = 0$

$f^{(5)}(x) = \cos x \qquad f^{(5)}(0) = 1$

$f^{(6)}(x) = -\sin x \qquad f^{(6)}(0) = 0$

$f^{(7)}(x) = -\cos x \qquad f^{(7)}(0) = -1$

∴ Maclauren series

$$f(x) = \frac{1}{1!}x^1 + \frac{(-1)}{3!}x^3 + \frac{1}{5!}x^5 + \frac{(-1)}{7!}x^7 + \dots$$

$$= \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

e) $f(x) = \tan^{-1} x$

$f'(x) = \frac{1}{1+x^2} = (1+x^2)^{-1} = y'$

$f''(x) = (-1)(1+x^2)^{-2}(2x)$

$= -2x(1+x^2)^{-2} = -2x(y')^2$

$f'''(x) = -2x(2)(y')(y'') + (-2)(y')^2$

$= 8x^2(y')^3 - 2(y')^2$

$\vdots$

Solution for Example 8 (cont.)

f)

$f(x) = \sinh x \qquad f(0) = 0$

$f'(x) = \cosh x \qquad f'(0) = 1$

$f''(x) = \sinh x \qquad \vdots$

$f'''(x) = \cosh x \qquad f'''(0) = 1$

$f^{(4)}(x) = \sinh x \qquad \vdots$

$f^{(5)}(x) = \cosh x \qquad f^{(5)}(0) = 1$

$f^{(6)}(x) = \sinh x \qquad \vdots$

$f^{(7)}(x) = \cosh x \qquad f^{(7)}(0) = 1$

$\therefore f(x) = \frac{1}{1!}x^1 + \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \frac{1}{7}x^7 + \dots$

h)

$f(x) = \tanh^{-1} x \qquad f(0) = 0$

$f'(x) = \frac{1}{1-x^2} = (1-x^2)^{-1} \qquad f'(0) = 1$

$f''(x) = -(1-x^2)^{-2}(-2x) \qquad f''(0) = 0$

$= 2x(1-x^2)^{-2} \qquad f'''(0) = 2$

$f'''(x) = 2x(-2)(1-x^2)^{-3}(-2x) + 2(1-x^2)^{-2} \qquad \vdots$

$= 8x^2(1-x)^{-3} + 2(1-x^2)^{-2}$

$f(x) = \frac{1}{1!}x^1 + \frac{2}{3!}x^3 + \dots$

$= x + \frac{x^3}{3} + \dots$

Example 9 (CIE 2020w21/1)

(a) By differentiating e^{-x^2}, find the Maclaurin's series for e^{-x^2} up to and including the term in x^2. [5]

(b) Deduce an approximation to $\int_0^{\frac{1}{5}} e^{-x^2}\,dx$, giving your answer as a rational fraction in its lowest terms. [2]

$\left(Ans\text{: (a) } 1 - x^2;\ \text{(b) } \frac{74}{375}\right)$

Solution

a) Let $f(x) = e^{-x^2}$ $\quad f(0) = 1$

$f'(x) = -2x(e^{-x^2})$ $\quad f'(0) = 0$

$f''(x) = -2x(-2x\,e^{-x^2}) + (-2)(e^{-x^2})$

$= 4x^2e^{-x^2} - 2e^{-x^2}$

$= (4x^2 - 2)e^{-x^2}$ $\quad f''(0) = -2$

$\therefore f(x) = 1 + \frac{0}{1!}x + \frac{(-2)}{2!}x^2 + \ldots$

$= 1 - x^2 + \ldots$

b) $\int_0^{1/5} e^{-x^2}\,dx \approx \int_0^{1/5} (1 - x^2)\,dx$

$= \left[(x - \frac{1}{3}x^3)\right]_0^{1/5}$

$= \left(\frac{1}{5} - \frac{1}{3}(\frac{1}{5})^3\right) - 0$

$= \frac{74}{375}$

Example 10

Using calculus, find the Maclaurin's series for $\tan x$, giving the first three non-zero terms.

$\left(Ans\text{: } \tan x \approx x + \frac{1}{3}x^3 + \frac{2}{15}x^5\right)$

[***Hint***: To make things simpler, use the identity $1 + \tan^2 x = \sec^2 x$ to write $y' = 1 + y^2$, then perform **implicit differentiation**.]

Solution

Let $y = \tan x$ $\quad y = 0$

$y' = \sec^2 x$ $\quad y' = 1$

$y'' = 2\sec x(\sec x \tan x)$

$= 2\sec^2 x \tan x$

$= [2(y')][(y)]$

$= 2y(y')$ $\quad \boxed{\sec^2 x = 1 + \tan^2 x}$

$= 2y(1 + y^2)$

$= 2y + 2y^3$ $\quad y'' = 0$

$y''' = 2y' + 6y^2y'$

$= 2(1+y^2) + 6y^2(1+y^2)$

$= 2 + 8y^2 + 6y^4$ $\quad y''' = 2$

$y^{(4)} = 16yy' + 24y^3y'$

$= 16y(1+y^2) + 24y^3(1+y^2)$

$= 16y + 40y^3 + 24y^5$ $\quad y^{(4)} = 0$

$y^{(5)} = 16y' + 120y^2y' + 120y^4y'$ $\quad y^{(5)} = 16$

$\tan x = 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 + \frac{2}{3!}x^3 + \frac{0}{4!}x^4 + \frac{16}{5!}x^5 + \ldots$

$= x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \ldots$